

TECHNICAL REPORT: Acoustic UAS Classification

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1. Introduction

The increase in use of unmanned aerial (drone) systems has created many airspace security problems, as seen in the 2018 Gatwick Airport shutdown and in the widespread commercial quadcopter use in the ongoing world conflict. Passive acoustic sensing is used alongside radar and radio frequency detectors since passive acoustic sensors are inexpensive, operate covertly, and do not need a direct line of sight or radio frequency emission from the target (Svanström et al., 2021). Studies investigating acoustic detection methods often report very high accuracy in specific recording environments (Casabianca & Zhang, 2021; Al-Emadi et al., 2019); however, there are three key aspects related to deploying these types of methods that could use a closer look: Do the models generalize to different recording conditions? Can models differentiate between a legitimate rotorcraft (such as a medical helicopter) and legitimate drones based upon acoustics alone? Will the models continue to perform when presented with lower levels of signal to noise ratio?

The purpose of this project is to investigate these questions. Acoustic UAS detection is framed as a 3 class problem, drone, helicopter, background, and three architectures are compared: a linear support vector machine using hand-crafted feature sets, a five-block convolutional network (CNN) trained from scratch on log-mel spectrograms, and a PANN's CNN10 architecture transferred using progressive unfreezing. Each model is tested on three different metrics: in-distribution testing, a cross-source generalisation holdout, and an SNR degradation curve.

2. Related Work

Previous research in acoustic detection of UAS and drones utilized Mel Frequency Cepstral Coefficients (MFCCs) with Gaussian Mixture Models (GMM) or shallow classifiers (e.g. Jeon et al., 2017). Since then, deep learning has dominated research in this area. Al-Emadi et al. (2019) published a large-scale publicly available drone audio dataset and investigated the performance of CNNs, RNNs, and CRNNs for the purpose of identifying binary drone presence. Later, Casabianca and Zhang (2021) expanded on previous comparisons conducted on MFCC-based CNNs utilizing mel-spectrogram inputs. They found that an average accuracy of 94.7% could be achieved through a late-fused ensemble model, with CNNs being superior to RNNs and CRNNs in terms of individual model performance. Additional research in the field includes hybrid CNN-Transformer models for acoustic drone identification (Jasim & Hreshee, 2025) and multi-modal fusion pipelines integrating visual information, LiDAR information, and radar information (Deng et al., 2024). However, a common limitation among most of this prior research is that evaluations were performed solely within a single recording source. Therefore, it remains unknown whether reported accuracy values will translate to new microphones, drone designs/models, or environmental settings.

Log-Mel Spectrograms as Inputs to Image-Based CNN Architectures. Using log-mel spectrograms as two-dimensional inputs to image-based CNN architectures was made popular by Hershey et al. (2017), they examined several pre-trained vision-based architectures applied to audio classification tasks on a 70M clip YouTube video database. The results indicated that architectures designed for images generally produced accurate results for audio classification tasks. This approach serves as the basis for the scratch CNN and PANN backbone used in this study.

Audio transfer learning. Kong et al. (2020) developed the PANNs family of models trained on nearly two million AudioSet clips. Through experiments they demonstrated that fine-tuning significantly improves performance relative to training from scratch on small amounts of data, e.g. improved ESC-50 accuracy from 0.833 to 0.947. Although effective fine-tuning on small data sets requires SpecAugment (Park et al., 2019) and MixUp (Kong et al., 2020) which we use, this on evaluating a PANNs CNN10 model against a matched scratch CNN model and a traditional classical baseline method under a unified experimental design involving in-distribution, cross-source, and SNR degradation axes.

3. Methodology

3.1 Datasets

Three public datasets are used for this study. Svanström's multi-sensor drone detection set (Svanström et al., 2021) provides the majority of the data for the study and consists of 90 ten-second long audio recordings at 44.1 kHz. The recordings consist of 30 recordings each of drones, helicopters, and backgrounds collected at three airports located in Sweden (Gothenburg City Airport, Malmö Airport, Halmstad Airport). The Al-Emadi DroneAudioDataset (Al-Emadi et al., 2019) is used in its entirety as a cross-domain holdout and consists of 11,704 files containing recordings of drones and backgrounds only. The Environmental Sound Classification Dataset ESC-50 (Piczak, 2015) is utilized as an environmental noise collection consisting of 320 environmental sounds including: wind, rain, crackling fire, crickets, thunderstorm, footsteps, car horn, engine, for generating an SNR degradation curve. All recordings are downsampled to 32 kHz mono format for consistency with the PANNs specifications (Kong et al., 2020), and divided into non-overlapping two second window segments with one second overlap. Due to potential overlap between windows during segmentation, all splits are made at the recording level while maintaining a consistent random seed to ensure no leakage occurs between train-validation-test splits resulting in 567 / 108 / 135 segment splits respectively.

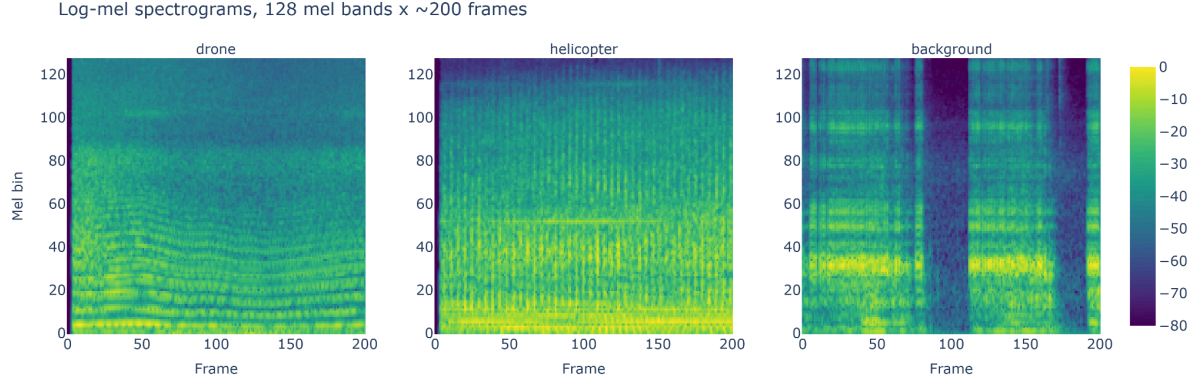


Figure 1: Example Spectrogram

3.2 Architecture

Three different architectures are compared. The first is a traditional classical baseline employing a linear Support Vector Machine (SVM) over a set of hand-crafted features: MFCC means and standard deviations (13 coefficients), spectral centroid, rolloff and bandwidth summaries, zero-crossing rate, and RMS energy. LinearSVC is used with balanced class weighting and wrapped in a CalibratedClassifierCV for probability; standardization is then applied in a scikit-learn pipeline.

The second architecture employs a CNN trained from scratch and utilizes five blocks of Conv2d BatchNormalization, ReLU, MaxPool layers with increasing number of channels (32, 64, 128, 256, 512) followed by an adaptive average pooling layer, a 256-unit fully connected layer, dropout ($p = 0.3$), and a three-way classifier, totaling 1.70M trainable parameters.

The third architecture utilizes the PANNs CNN10 architecture (Kong et al., 2020) which uses four double-convolution blocks with channels (64, 128, 256, 512), global mean-over-time and max-over-mel pooling layers, a 512-unit embedding layer, dropout, and a three-way classifier, totaling 4.95M trainable parameters with pre-trained AudioSet weights.

3.3 Training

Each neural network is trained using Adam optimizer with L2 regularization at 1×10^{-4} and class weighted cross entropy loss function. Additionally, both log-mel spectrogram inputs are subject to SpecAugment (two time masks with parameter 30, two frequency masks with parameter 20), following Park et al. (2019), and MixUp ($\alpha = 0.2$) following Kong et al. (2020). Early Stopping tracks validation macro F1-score with maximum patience of 15 for both the scratch CNN architecture and 8 per PANN stage. Progressive unfreezing is implemented across three stages for PANN fine-tuning: head only (10 epochs with learning rate 1×10^{-3} ; total trainable parameters 264k); last convolutional block (15 epochs; head learning rate 1×10^{-3} ; backbone learning rate 1×10^{-5} ; total trainable parameters 3.81M); last two blocks (20 epochs, same schedule, parameters 4.69M).

3.4 Evaluation

The three evaluation criteria use the same method: macro F1 with 95% bootstrap confidence interval (1,000 resamples on a fixed seed) and per class one vs rest AUC ROC. The macro F1 will average over only those classes that have been used in the previous holdout. Therefore the missing class will not reduce the score. The three evaluation criteria are in distribution testing of the Svanstrom test set, cross source evaluation on the Al-Emadi hold-out (drone and background only) , and an SNR degradation curve at clean (0dB), +10dB, +5dB, 0dB, -5dB using ESC-50 as the additive noise source (drone and helicopter only, background is environmental audio so it cannot have a SNR).

4. Results

NOTE: The numbers may change on new runs.

4.1 Performance

All three models have good in-distribution results on the Svanstrom test set. PANN CNN10 performed best with macro-F1 = 0.932 [0.886, 0.970], followed by the scratch CNN with macro-F1 = 0.910 [0.857, 0.953] and the SVM with macro-F1 = 0.874 [0.806, 0.925]. All three models had AUC-ROC > 0.99 for both drone and helicopter classes for both neural models. The bootstrap confidence intervals for CNN and PANN were very close therefore the pre-training advantages could be real but are not statistically significant based on 135 test examples.

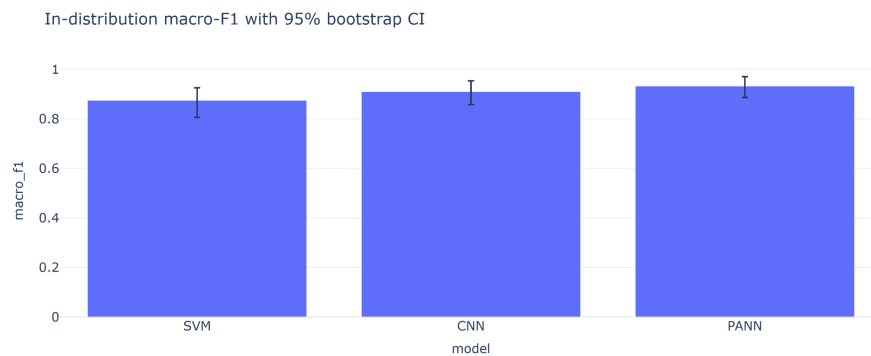


Figure 2: In Distribution

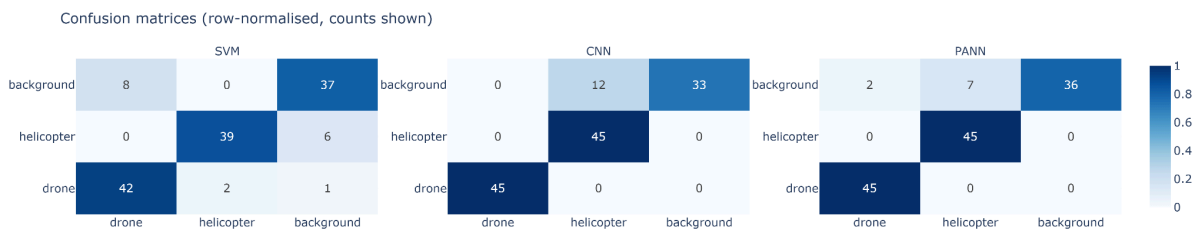


Figure 3: Confusion Matrices

The confusion matrices show that both neural models classify every drone and helicopter segment perfectly (45/45 each), and therefore the errors in classification of each network are only due to background being incorrectly classified as either helicopter or drone. The CNN incorrectly classified 12 of 45 background segments as helicopter while the PANN incorrectly classified 7 of 45 background segments as helicopter. Therefore the first part of the classification problem (whether you tell apart a drone from another type of rotor craft based solely upon the audio) was solved successfully for all three networks. However, the second part of the classification problem (whether you distinguish between machine noise and non-machine noise based upon the audio) is still open.



Figure 4: CNN Curves



Figure 5: PANN Curves

The CNN training history has an unstable validation loss early in the training but stable validation loss after (epoch 13) with early stopping at epoch 28, the PANN training history showed a continuous improvement, the validation macro-F1 rose from approximately 0.78 at the end of head-only training to 0.85 after the last block was unfrozen and then to 0.88 after the last two blocks were unfrozen. These results are in favour of progressive unfreezing of weights during transfer learning rather than a single-stage unfreeze.

4.2 Generalisation

On the AI-Emadi holdout all three networks dropped in performance but the order of the networks remained consistent. The SVM dropped to 0.426 (-51%) . The scratch CNN dropped to 0.518 (-43%) . The PANN dropped to 0.568 (-39%). The richer training signal resulted in a lower, albeit still sizable, domain shift penalty.

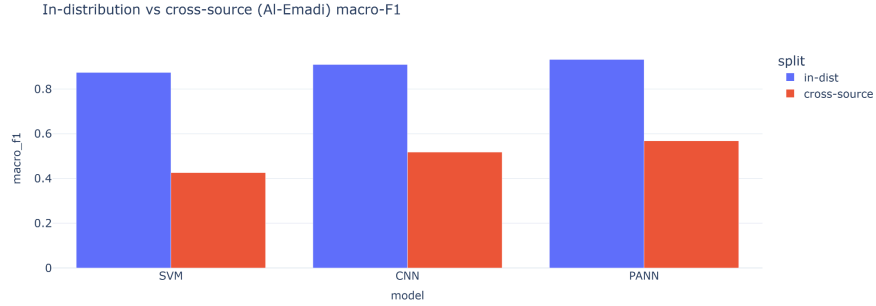


Figure 6: Cross Source

4.3 SNR degradation

Restricted to only drones and helicopters, all three networks degraded smoothly as the amount of noise increased. Both neural nets achieved a macro-F1 = 1.00 at clean conditions; by -5 dB the scratch CNN (0.834) slightly edged the PANN (0.810), reversing the order seen in-distribution. The SVM degraded faster than any other network dropping from 0.936 to 0.793.

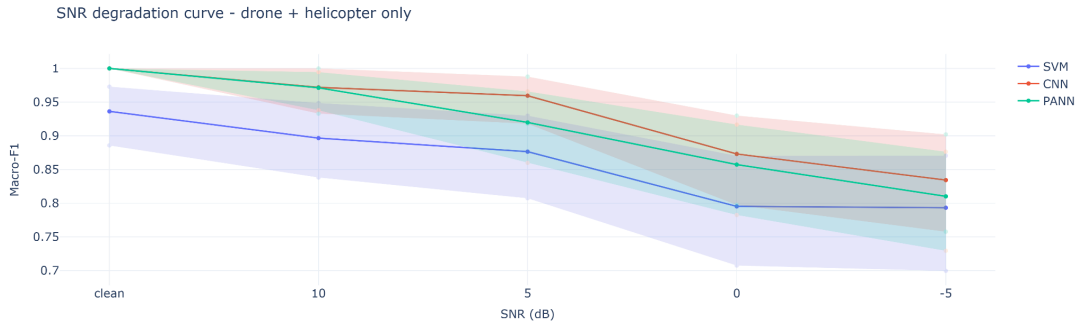


Figure 7: SNR Curve

4.4 Summary

Therefore no single network dominated all of the evaluations. The PANN was clearly superior in-distribution and showed a greater degree of robustness to cross-source shift than did either of the other two networks. The scratch CNN achieved or nearly achieved equal performance to that of the PANN under high levels of noise; and the SVM demonstrated significantly weaker performance than both neural networks, yet clearly demonstrated that both of these architectures achieved performance far better than random guessing. Architecture choice depends upon which deployment angle is operationally most relevant.

5. Conclusions

The cross source axis was most relevant. Although the PANN had the smallest drop, all three models suffered between a 39-51% loss in macro-F1, suggesting that pre-training provides some mitigation of domain shift resulting from differences in microphone, drone platform, and environment, but not total mitigation, so domain fine-tuning or explicit domain adaptation are likely to still be required if pre-training is used as the only technique for deploying to a new airport or sensor setup.

The CNN did better than PANN at -5dB, flipping the in-distribution order. This may be due to the fact that AudioSet clips are mostly clean recordings, and the PANN has likely learnt features that aren't as transferable to noisy spectrograms, while the scratch CNN has learnt noisy features on Svanström. So it seems "transfer learning always wins" is not necessarily true, when choosing an architecture we might want to consider expected noise.

For the link back to module content; the three models implement almost all the modules: soft-margin classification with class-weighting (Week 2); dropout, weight decay, early-stopping and data-augmentation to regularise (Week 4); convolutional hierarchical features, batch normalization and global-pooling (Week 5); and progressively unfreezing weights for transfer learning with a pre-trained backbone.

There are several limitations though. The main dataset is small (around 90 recordings). The helicopter class is missing from the cross source holdout so there is no test for domain shift with drone vs helicopter, and SNR is simulated by adding ESC-50 noise, rather than measured on noisy recordings.

For future work we will explore more cross source rotorcraft datasets, which would increase the generalisability of the results, domain adaptation techniques, e.g. DANN or CORAL, real-world outdoor noise recordings and waveform input using Wavegram-Logmel-CNN model (Kong et al., 2020).

6. References

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